



**Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)**

Experiment 6

ABSORBANCE CURVES OF TWO COLORED COMPOUNDS (APPLICATION OF BEER'S LAW)

Adopted from: B 22 C MLAB 2401 - Clinical Chemistry Lab Manual*

Pre-Lab Questions:

1. The reagent blank corrects for the absorbance caused by what?
2. To establish a standard curve for a chemical, the spectrophotometer wavelength is set on which value?
3. Using a standard curve, if you know the absorbance of an unknown sample, what else can be determined about the unknown?
4. What is the function of the photodetector in the spectrophotometer?

Precautions:

1. In this experiment we aim to calculate the exact concentration of two colored compounds. Make sure to use the accurate measurement equipment (avoid using transfer pipettes)
2. Make sure to use clean cuvettes and wipe their outside from any residual material or any fingertips

Objectives:

1. Use proper technique in operating a spectrophotometer and make accurate readings on absorbance curves.
2. Quantitation of a mixture of two colored compounds (organic dyes) by the application Of Beer's Law.
2. determine lambda max for each one of the specified dyes
3. preparation of standard solutions each with a specified concentration
3. Plot the standard curve (absorbance curve) for two colored compounds
4. Use the standard curve to calculate concentration of two colored compounds

Introduction:

Spectrophotometry involves the use of a spectrophotometer which measures the light intensity as a function of color or more generally the wave length of the light. There are different types of spectrophotometers depending on the wavelengths they are supposed to work with (depending source of light they use).

Spectrum is defined as the range of wavelengths characteristic of a specific type of radiation check (figure one).

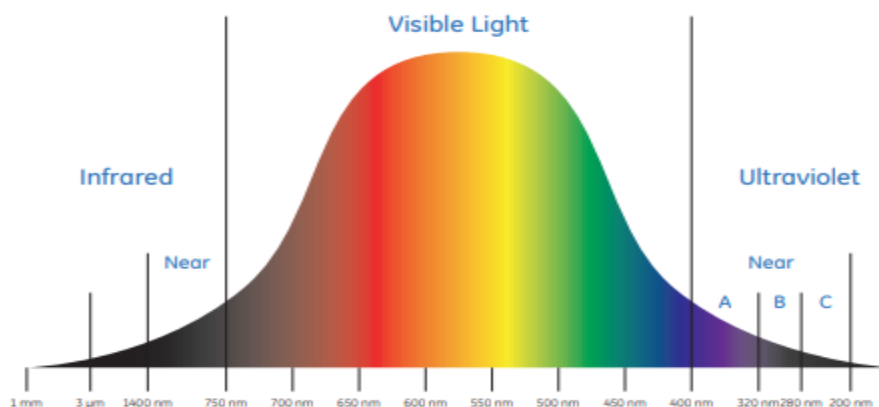


Figure one : The electromagnetic spectrum which most of the spectrophotometers operate in the infrared, visible, UV sections. All the test we will run in our biochemistry lab will use the visible light area on the electromagnetic spectrum.

https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/Sigma-Aldrich/General_Information/1/ge-spectrophotometry.pdf

Photometry (direct colorimetry) measures absorbance (A) also called optical density (OD), or percent transmittance (%T). The depth of the solution (light path) is held constant using a standard cuvette. (Properties of cuvettes are critical.) (figure 2). As the absorbance of a solution increases, %T decreases logarithmically. The radiation not transmitted is absorbed depending on the characteristic of a given molecule which is called molar absorptivity.

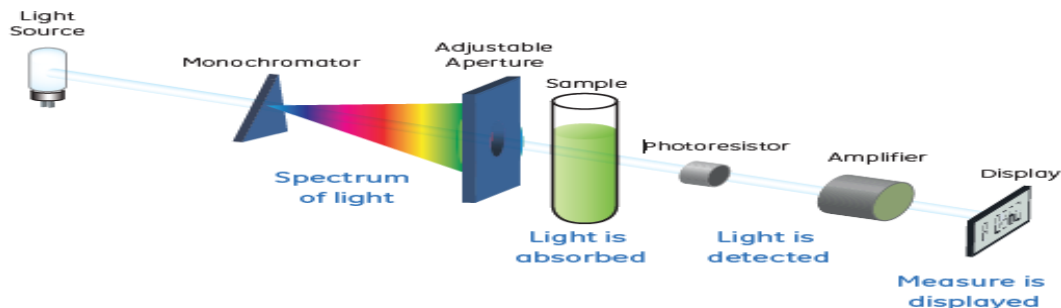


Figure two: schematic view of the spectrophotometer parts. A spectrophotometer is an instrument used to measure the absorbance and some other functions

https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/Sigma-Aldrich/General_Information/1/ge-spectrophotometry.pdf

The absorption and/or transmission of light through a specimen is used to determine molar concentration of a substance. The relationships between absorption, transmittance and concentration is stated using **the Beer Lambert Law (figure3)**

Which states that "the amount of light absorbed is proportional to the concentration of a solution" in other words there is a linear relationship between the absorbance and concentration (**figure 4**)

Beer's law is widely applied in the field of analytical and biochemistry specially in biomedical research and medicine.

$$A = a * b * C$$

absorbance
molar absorptivity constant (depends on combination of solute/solvent/ λ)
length of path traveled by light
solution concentration (mol/L)

Figure(3) the Beers Lambert law **a** is given by $\text{mol L}^{-1} \text{cm}^{-1}$. **b** is equal 1.0 cm. and **C** in terms of mol/L

the main disadvantage of Beer's law is the deviation from linearity between absorbance and concentration at very high concentrations of the analyte due to the electrostatic interactions between molecules in close proximity scattering of light
Deviation can simply be minimized by diluting solutions and the result is multiplied by dilution factor.

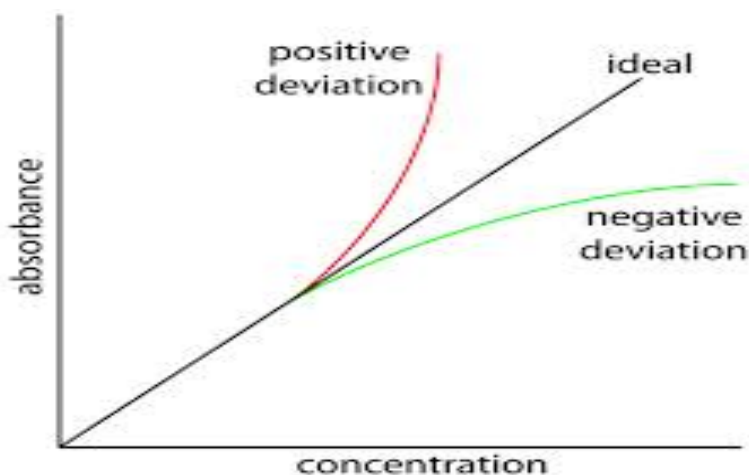


Figure 4 Beer's Law: The absorbance of light is directly proportional to both the concentration of the absorbing medium (represented by a linear line) to a certain concentration. At a very high concentration of the absorbing medium deviation from linearity could occur.

Absorbance –Wavelength diagram to determine $\lambda_{\max}$

Lambda max ($\lambda_{\max}$) refers to the wavelength in the light spectrum characterized with the maximum absorbance value. Generally molecules absorb in a wavelength range centered around the lambda max (figure 5) (table1).

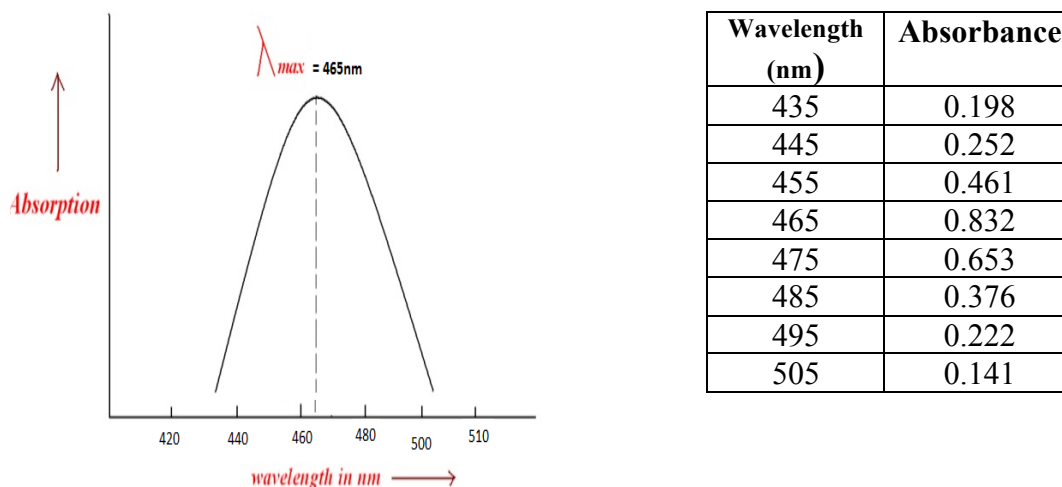


Figure 5 the curve of $\lambda_{\max}$ determination for organic dye (methyl orange). the X-axis represents a scale of wavelengths and the Y-axis represents a scale for absorbance values, the values obtained by experiment shown in (table one) are plotted to generate the curve

($\lambda_{\max}$) for Methyl orange (MO) is 465 nm and that for Bromophenol blue (BPB) is 595 nm. Since there is no interference between their wavelengths as shown in the following figure 5, we can measure absorbance of each one of them in a mixture containing both without any dye interfering with the absorbance of the other

This can be done adjusting the spectrophotometer at the wavelength specific for each dye when the spectrophotometry is set at wavelength of 465 nm we will only measure the absorbance of Methyl Orange when the wavelength is adjusted at 595 nm on the spectrophotometer one can measure the absorbance of Bromophenol Blue only from the mixture.

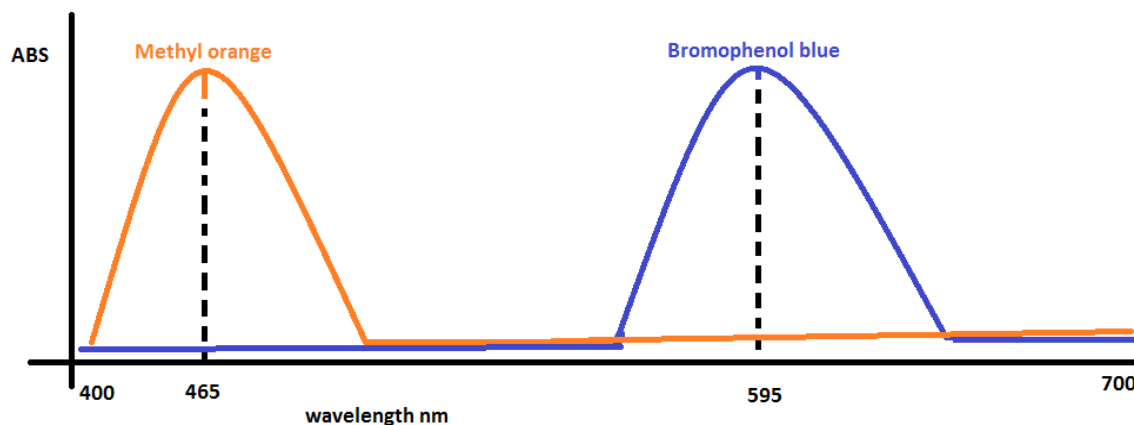


figure 5 absorbance curves of two colored compounds: methyl orange and Bromophenol blue

Procedure:

1- Prepare the following two sets of standards using a pipit to have accurate volumes.

A) Methyl Orange standards (The original concentration of MO stock solution is 10.0 mg/L).

Solution	1*	2	3	4	5
Volume of MO (mL)	1.0	2.0	3.0	4.0	5.0
Volume of water (mL)	4.0	3.0	2.0	1.0	0.0

B) Bromophenol Blue standards (The original concentration of Bromophenol Blue stock solution is 10.0 mg/L).

Solution	1*	2	3	4	5
Volume of BPB (mL)	1.0	2.0	3.0	4.0	5.0
Volume of water (mL)	4.0	3.0	2.0	1.0	0.0

2- On the spectrophotometer for at least ten minutes to warm up then adjust the proper wave length (at 465 nm for MO)

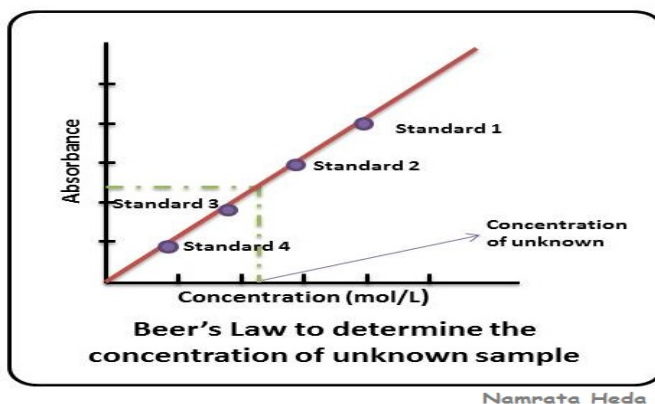
3- Prepare the blank solution* and adjust the spectrophotometer at zero absorbance.

4- Measure the absorbance of each standard and tabulate your results.

5- Construct a calibration curve by drawing the relationship between absorbance and concentration.

6- Measure the absorbance of the solution mixture which refers to MO only and determine the concentration of the unknown sample.

7- Repeat the above steps for BPB at a wave length 595 nm and determine the concentration of BPB in the unknown sample.



*A blank solution is used to zeroing the spectrophotometer or re zeroing it if the wave length is changed. It consists of all materials except the substance under investigation.

References

- 1- http://employees.oneonta.edu/kotzjc/LAB/Spec_intro.pdf
- 2- https://www.sigmaaldrich.com/content/dam/sigma-aldrich/docs/Sigma-Aldrich/General_Information/1/ge-spectrophotometry.pdf
- 3- <https://msu.edu/course/lbs/159h/Spectrophotometry04.pdf>



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Applied Chemistry
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Experiment 6**

**ABSORBANCE CURVES OF TWO COLORED COMPOUNDS
(APPLICATION OF BEER'S LAW)**

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Unknown #:

Unknown Identification:

Purpose:

Data and calculations:

On a graph paper plot absorbance (A) of the dye versus its concentration (C).

Concentration (mg /L)	Absorbance of Methyl Orange (MO)	Absorbance of Bromophenol Blue (BPB)
2		
4		
6		
8		
10		
Unknown #,		
Unknown #,		

From the curves

The concentration of Methyl Orange in the first unknown is:

The concentration of Methyl Orange in the first unknown is:

The concentration of Bromophenol Blue in the first unknown is:

The concentration of Bromophenol Blue in the second unknown is:

Conclusion:

Errors encountered running the experiment